

Element A

Presentation and justification of the problem

The problem is clearly and objectively identified and defined with considerable depth, and it is well elaborated with specific detail; the justification of the problem highlights the concerns of many primary stakeholders and is based on comprehensive, timely, and consistently credible sources; it offers consistently objective detail from which multiple measurable design requirements can be determined. The Andean Bears at the Nashville Zoo are very intelligent creatures, equivalent to a Belgian Malinois, which, according to scientists at Helsinki University is the smartest dog breed, totalling 35 points out of 39, a whole 9 points over the second placed Border Collie. Day after day the two bears, Pinoccio and Luka are confined to the same areas of the zoo. A scenic enclosure which allows them to roam free and catch the eyes of interested visitors, and a cage back behind the enclosure for other times. The bears are only allowed in the enclosure with each other 8 months of the year due to mating season. Without a companion and proper mental stimulation the bears are at risk of developing depression and anxiety as their mental health deteriorates.

We, the Brentwood High School Engineering class, have been tasked by the Andean Bear caretakers to create an engaging puzzle-like toy that will keep them entertained during their long repetitive days at the zoo. This task comes with several challenges that need to be worked around, including but not limited to, strength, difficulty, reusability, and safety. An important element of the project is making sure that it is stimulating, but not too complicated where they aren't able to solve it. Leading us to further investigate the intelligence of the Andean Bear, and their capacity for problem solving. Through this research we found that the bears are smart enough to use basic tools, leading us to believe it would be reasonable for us to assume that the

bears would be able to learn to use levers/buttons to activate various puzzles or devices. The keepers have also requested that, “The longer the duration the better. We prefer our bears to be engaged with the item as long as possible and it's always great for them to return to it during different periods throughout the day.” The challenge of making something mentally stimulating is further made difficult by the need to keep the puzzle strong enough to withstand the bears for a long period of time. According to initial questions asked of the zookeepers, we were told that the lowest level PVC pipe used for the bears is schedule 40, and the recommended was schedule 80. Using this information we were able to standardize the amount of force exerted by the bears as well as finding a floor for our materials strength wise. Lastly, we were tasked with ensuring the safety of the bears, zookeepers and our group as we manufactured the puzzle. This led us to considering, toxicity, flammability, combustability, and sharpness.





- 1.) How often do the handlers enter the enclosure? **Keeper staff will enter the habitat in the AM and have the ability to access in the afternoon if needed . As for their inside dens, they can access multiple times/day as needed.**
- 2.) How long do the bears need to be entertained (short and long-term)? **The longer the duration the better. We prefer our bears to be engaged with the item as long as possible and its always great for them to return to it during different periods throughout the day.**

- 3.) Would you like the puzzle to be permanent or temporarily in the cage? **We would like the enrichment device to be temporary. We would like to have the ability to move it to different places within the habitat as well as to their inside dens.**
- 4.) How do the handlers expect to administer the puzzle/extract the puzzle? **The keepers will load the enrichment device with their diet and move it to the habitat or dens by themselves by hand.**
- 5.) What materials are safe for the bears/not safe? **Bears are extremely strong animals so if you are making something from PVC, I would concentrate on schedule 40 or schedule 80 PVC. This material would be durable enough for them and easy to clean. Thick wood, 4x4, can also be used but it is not an easy to clean.**
- 6.) What materials are preferred? **See above**
- 7.) Do the puzzles need to be food-based (what incentivizes the bears other than food)? **Bears are extremely food motivated so a majority of their enrichment devices revolve around food. They also like to push on items such as large car/truck tires.**
- 8.) Do the bears learn habits, if so how fast? **Yes, they do learn habits and can develop one as quickly as a day.**
- 9.) Are we allowed to modify currently allocated resources (the current purple geometric toy inside the external cage for example)? **We can provide you with this to modify if necessary.**
- 10.) What size should the largest portion of food be to fit inside the puzzle? We provided dry food items to the class when you were here last but the largest size that would be used would range from 2"x2" to 3"
- 11.) In terms of dimensions, we decided that the maximum size of any possible design of the puzzle feeder is the size of a 31-gallon water drum. (22 inches in diameter, 35 inches tall). **Is this a reasonable size? Yes! that's perfect!**
- 12.) How long are the Andean bears' attention span? **It's all dependent on the bear and the day. If they are foraging for food via an enrichment device, they could be engaged for 5-10minutes or until the food is depleted.**
- 13.) For reference, the engineering class currently has 3 groups working separately to build the puzzle feeders. In preference to the zookeepers, how many groups should be

working mounted feeder instead of a non-mounted feeder? **All feeder devices that will be on habitat will need to be mounted one way or another. We will use either a chain or an eye bolt to secure. This is to ensure that the items do not make it into the moat, pool or hit the glass. Inside their dens, they are able to have an unmounted as well as a mounted device.**

- 14.) Can the feeder become detached from another portion of the feeder while inside the enclosure? **We can not have the feeder become detached if it is on the exhibit.**

<https://www.countryliving.com/uk/wildlife/dog-breeds/a42429920/intelligent-dog-breeds/>

Proof of intelligence using tools

<https://science.sandiegozoo.org/science-blog/even-well-fed-andean-bears-creatively-try-get-goodies>

Requirement of mental stimulus study

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8908830/>